Curriculum vitae

Ankit Singh

CONTACT INFORMATION

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RESEARCH INTERESTS

Galaxy evolution, large-scale environment, hydrodynamical simulations, Active Galactic Nuclei (AGN).

EXPERIENCE

University of Nottingham, United Kingdom

Research Fellow (with Prof. Meghan Gray and Prof. Frazer Pearce) 2025–2026

Korea Institute for Advanced Study (KIAS), Republic of Korea

Research Fellow (with Prof. Changbom Park) 2020–2024

EDUCATION

Indian Institute of Science Education and Research (IISER Mohali), India

PhD in Physics 2014–2020

Supervisor: Prof. Jasjeet Singh Bagla; **Thesis title**: Galaxies and their environment

Department of Physics and Astrophysics, University of Delhi

MSc in Physics 2012–2014

ACADEMIC SERVICE

Referee for the Astrophysical Journal, 2021–2024

TALKS AND POSTERS

- Talk on *A Scalable Frozen-Core Approach to Topological Filament Finding in Large Cosmological Simulations* at the National Astronomy Meeting, 20–24 July 2026.
- Talk on *Calculating Dust-attenuated SEDs for galaxies in Horizon Run 5 Simulation using the Powderday Code* at Challenges and Innovations in Computational Astrophysics VI at IISER Mohali, India, 29 September–2 October 2025.
- Talk on *Probing the chemical evolution of elliptical galaxies within galaxy clusters with HR5* at the HR5 collaboration meeting 2024, KIAS, 4 November 2024.
- Talk on *How does the local environment affect AGN activity in a galaxy?* at the 11th KIAS Workshop on Cosmology and Structure Formation at Gyeongju, Korea, 27 October–2 November 2024.
- Poster on *Effects of local environment on AGN activity* in GasPisa2024: The physical processes shaping the stellar and gaseous histories of galaxies at Pisa, Italy, 27–31 May 2024.
- Talk on *Study of the effect of environment on AGN activity in HR5 simulation* at the University of Hull, UK, 15 May 2023.
- Online astrophysics outreach talk on *Studying Galaxy Evolution using Hydrodynamical Simulations* at Nehru Planetarium, New Delhi, 16 February 2023.
- Talk on *Impact of AGN emission on estimated star formation rate using IR luminosity of galaxies in HR5* at Mapping the Invisible Universe, Lorentz Center, Leiden, The Netherlands, 29 August–2 September 2022.

- Talk on *Effects of local environment on AGN activity in HR5* at the 10th KIAS Workshop on Cosmology and Structure Formation at KIAS, 24–28 October 2022.
- Talk on *Study of filament galaxies in EAGLE simulation*, *LSST@ASIA: Exploring the Wide, Fast, Deep Universe* conference at the University of New South Wales (UNSW), Sydney, Australia, 20–23 May 2019.
- Talk on *Study of filament galaxies in EAGLE simulation*, XXXVII Meeting of Astronomical Society of India, Christ (Deemed to be University), Bengaluru, India, 18–22 February 2019.
- Poster on *Caught in the web: a tale of filament galaxies* in IAU Symposium 341: PanModel2018: Challenges in panchromatic modelling of galaxies with next-generation facilities, Osaka University, Osaka, Japan, 12–16 November 2018.
- Talk on *Ultraviolet and optical view of galaxies in Coma Supercluster*, Young Astronomer's Meet (YAM), Physical Research Laboratory (PRL), Ahmedabad, India, 24–28 September 2018.
- Talk on *An analytical study of ram pressure stripping of galaxies*, Franco-Indian Astronomy School: From Re-ionization to Large-Scale Structure: A Multi-wavelength Approach, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India, 11–17 February 2018.

WORKSHOPS

- Attended *Recent Advances in Galaxy Formation and Reionization* at Yonsei University, Seoul, South Korea, 6–10 November 2022.
- Workshop on *High-Performance Scientific Computing* at IISER Thiruvananthapuram, India, 6–10 June 2016.
- Training programme on *Linux and Linux system administration* (duration 5 days) and a follow-up training programme on *Network services and administration* (received A+ grade), IISER Mohali, 15 February–1 March 2016.
- National workshop on *Cosmology with HI 21cm line*, Raman Research Institute (RRI), Bengaluru, India, 23–26 June 2015.
- Fourteenth Workshop on *Instrumentation for Planetary Exploration*, D.D. University, Nadiad, Gujarat, India, 6–10 January 2014.
- BITS-IUCAA Workshop on *Gravitational Wave Data Analysis* at the Birla Institute of Technology and Science (BITS) Pilani, Goa campus, India, 17–21 December 2012.
- Workshop on *Numerical Methods and Differential Equations in Computational Science*, University of Delhi, 1–5 February 2011.

PUBLICATIONS

[1] **Ankit Singh**, Frazer Pearce, Meghan Gray, Ulrike Kuchner, Julian Onions, Gustavo Yepes, Alexander Knebe, "Channelled versus Isotropic Accretion of Galaxy Groups onto Clusters: Kinematics, Spin Alignment, and Assembly Bias", 2026, *MNRAS, accepted*

[2] **Ankit Singh**, Frazer R. Pearce, Meghan E. Gray, Gustavo Yepes, "Enabling Cosmic Web Analysis at Gigaparsec Scales: A Multi Block Approach for DisPerSE", 2026, *MNRAS, 550, stag1323*

[3] Aaron R. Rowntree, Fiorenzo Vincenzo, **Ankit Singh**, Gareth Few, Jaehyun Lee, Christophe Pichon, "The influence of galaxy mergers, black-hole growth, and gas processes on the evolution of the stellar mass-gas metallicity relation of galaxies in different cosmic environments", 2026, *MNRAS, stag1686*.

[4] Jaewon Yoo, Ellen Sirks, Cristiano Sabiu, Changbom Park, Jaehyun Lee, **Ankit Singh** et al., "Intracluster Light as a Probe for Dark Matter: Exploring Self-interacting Dark Matter and Cold Dark Matter with C-EAGLE Sims", 2026, *ApJ, 1008, 193*.

[5] Aaron R. Rowntree, Fiorenzo Vincenzo, **Ankit Singh**, Changbom Park, Jaehyun Lee, Christophe Pichon, Yohan Dubois, Gareth Few, Brad Gibson, Owain Snaith, Yonghwi Kim, "The Role of Large-Scale Environment in Shaping the Stellar Mass-Gas Metallicity Relation Across Time", 2026, *MNRAS, 546, stag199*.

[6] Aaron R. Rowntree, **Ankit Singh**, Fiorenzo Vincenzo, Brad K. Gibson, Céline Gouin, Daniela Galárraga-Espinosa, Jaehyun Lee, Juhan Kim, Clotilde Laigle, Changbom Park, Christophe Pichon, Gareth Few, Sungwook E. Hong, Yonghwi Kim, "The Environmental Dependence of the Stellar Mass - Gas Metallicity Relation in Horizon Run 5", 2024, *MNRAS, 531:3858–3875*.

[7] **Ankit Singh**, Shreya Davessar, Mamta Gulati, Jasjeet Singh Bagla, Meenu Prajapati, "Study of dependence of ram pressure stripping on the orbital parameters of the galaxies", 2024, *MNRAS, 530:699–709*

[8] Jaewon Yoo, Changbom Park, Cristiano Sabiu, **Ankit Singh**, Jongwan Ko, Jaehyun Lee, Christophe Pichon, M. James Jee, Brad K. Gibson, Owain Snaith, Juhan Kim, Jihye Shin, Yonghwi Kim, Hyowon Kim, "Spatial Distribution of Intracluster Light versus Dark Matter in Horizon Run 5", 2024, *ApJ, 965, 145*

[9] **Ankit Singh**, Changbom Park, Ena Choi, Juhan Kim, Hyunsung Jun, Brad K. Gibson, Yonghwi Kim, Jaehyun Lee, Owain Snaith, "On the Effects of Local Environment on Active Galactic Nucleus (AGN) in the Horizon Run 5 Simulation", 2023, *ApJ, 953, 64.*

[10] **Ankit Singh**, Smriti Mahajan, and Jasjeet Singh Bagla, "Study of galaxies on large-scale filaments in simulations", 2020, *MNRAS, 497(2):2265–2275*

[11] **Ankit Singh**, Mamta Gulati and Jasjeet Singh Bagla, "Ram pressure stripping: An analytical approach", 2019, *MNRAS, 489:5582–5593*

[12] **Ankit Singh**, Smriti Mahajan and Devika Shobhana "Caught in the web: A tale of filament galaxies", 2019, *Proceedings of the International Astronomical Union, 15(S341):304–306*

[13] Smriti Mahajan, **Ankit Singh** and Devika Shobhana "Ultraviolet and optical view of galaxies in the Coma supercluster", 2018, *MNRAS, 478:4336–4347*

AWARDS/ RECOGNITION

- Registration fee waiver for *The physical processes shaping the stellar and gaseous histories of galaxies* conference at Pisa, Italy, 27–31 May 2024.
- Registration fee waiver (AUD 750) and accommodation support (AUD 250) for attending *LSST@ASIA: Exploring the Wide, Fast, Deep Universe* conference at the University of New South Wales (UNSW), Sydney, Australia, 20–23 May 2019.
- Registration fee waiver and travel support for presenting in *XXXVII Meeting of Astronomical Society of India*, 18–22 February 2019.
- Travel support for attending *Franco-Indian Astronomy School*, IUCAA, Pune, India, 11–17 February 2018.
- Awarded Certificate of Merit (top 1% marks) in Science & Technology subject in secondary school examination by Central Board of Secondary Education (CBSE) in 2007.

TEACHING EXPERIENCE

- Lab tutor in the *Astrophysics* course in Semester II in 2017–18.
- Teaching assistant for *Numerical Methods* course in Semester II in 2014–15 and 2016–17.
- Teaching assistant for *Electricity and Magnetism lab* at IISER Mohali in Semester II in 2014–15.

- Teaching assistant for the Nuclear Physics lab at IISER Mohali for Semester I in years 2014–15, 2015–16 and 2016–17.

Public Engagement

- Organised sky-watching events in the Chandigarh Tricity area on 10 June 2017, 2 February, and 28 October 2018 in collaboration with the Society for Promotion of Science & Technology, India.
- Invited as a grader in the 10th International Olympiad on Astronomy and Astrophysics (IOAA), Bhubaneswar, India, 9–19 December 2016.

Organisation

- Local Organising Committee (LOC) member for the 11th KIAS Workshop on Cosmology and Structure Formation held in Gyeongju, Korea, 27 October–2 November 2024.
- LOC member for HR5 collaboration meetings on 24 August 2021 and 4 November 2024.
- LOC member for the 10th KIAS Workshop on Cosmology and Structure Formation held at KIAS, 24–28 October 2022.
- Volunteer at the International Conference on Gravitation and Cosmology (ICGC), IISER Mohali, 14–18 December 2016.

Computational Skills

- Proficient in Python for scientific analysis; working knowledge of Fortran and C.
- Machine learning (ML) and deep learning: PyTorch, convolutional neural networks, transfer learning, and computer vision for astronomical image classification.
- Experience in High-Performance Computing (HPC) management and usage.
- Relational databases, version control (git), and reproducible ML tooling (MLflow, Docker, continuous integration).

Certifications

- WorldQuant University, Applied AI Lab: Deep Learning for Computer Vision, and Applied Data Science Program, July 2025.
- Artificial Intelligence Bootcamp, OppWorks (hands-on modules in computer vision, natural language processing, and reinforcement learning), January 2026.
- AI+ Foundation, AI CERTs, October 2025.
- ISO/IEC 42001 Practitioner: AI Management Systems (AIMS), TRECCERT, September 2025.